LiTFSI 20 molal — provisional DLS/SLS analysis report

Executive Summary

Dataset. Water-in-salt electrolyte (LiTFSI, 20 mol/kg) by multi-angle PCS at 22 °C and 50 °C: 28 angles (15–150°), 5 replicates each, 3D-modulated cross-correlation (no polarization channels).

Main result. A broadly polydisperse / aggregate-laden electrolyte: a main population of a few hundred nm (z-average $R_h \approx 212$ nm at 22 °C, 85 nm at 50 °C) with a slow large-aggregate tail. The data sit in the **Porod regime** at every angle with a slope $I \propto q^{-3.2}$, with the sphere form factor in the PSD assumed, not confirmed; $\rho = R_g/R_h$ **is not determined** (no Guinier regime). The recorded viscosity (fixed 34 mPa·s at both T) is a placeholder; the same-sample solve implies $\eta(50\ °C) \approx 14$ mPa·s.

Confidence. The z-average r_h carries replicate/angle bootstrap intervals; leave-one-angle-out cv gives predictive r^2=0.985; the corenn inversion converged.

Assumption-sensitive. Viscosity, refractive index, q-from-angle calibration, the toluene/absolute-intensity reference, the assumed sphere form factor, and the CORENN regularization all enter the reported sizes; see the Assumption Register.

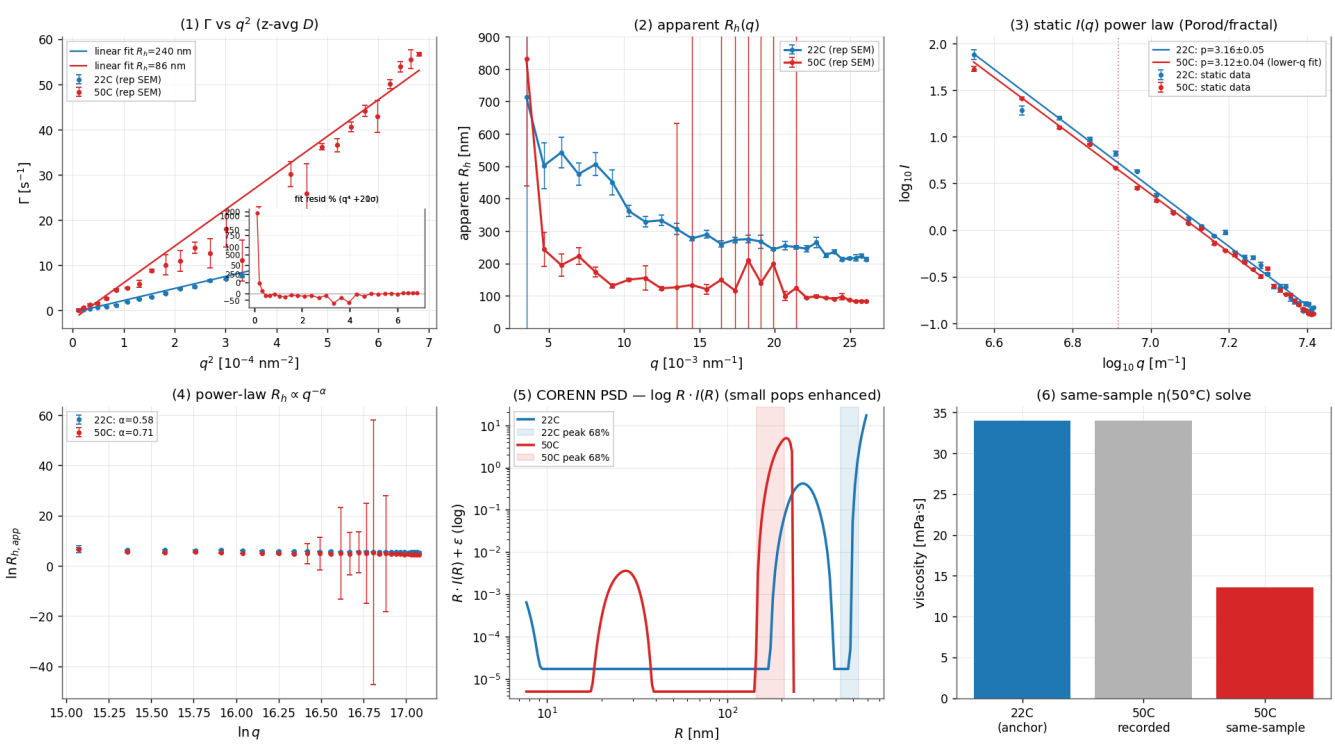
Recommended next action. measure the true η(T) of the electrolyte to convert the dynamics to an absolute size; fit a surface-/mass-fractal form factor (the Porod slope rules out a smooth sphere); confirm the toluene reference to lock the SLS scale.

> **Provisional flags:** viscosity is a fixed placeholder (corrected by same-sample solve); toluene/absolute-intensity reference not confirmable from the export; Porod slope $q^{-3.2}$ implies a surface-fractal / branched aggregate, not the sphere assumed by the PSD.

Summary Table

quantity	22 °C	50 °C
z-average R_h [68% CI]	212 [206–217] nm	85 [81–89] nm
shape $\rho = R_g/R_h$	not determined (Porod regime)	4.31 (rod / stiff / very extended (>1.9))
apparent R_h low→high q	506→215 nm	222→84 nm
PDI (mean)	0.30	0.68
static power-law $I \propto q^{-p}$	p=3.2	p=3.1
$\Gamma(q^2)$ predictive R^2 (LOO-CV)	0.985	0.926
CORENN eff.DOF / cond	19/24, κ=1e+14	10/24, κ=5e+09
power-law α	0.58	0.71
CORENN multi-angle PSD	530 nm	203 nm
CORENN peak [68% CI, bootstrap]	522 [424–538] nm	193 [145–208] nm
implied η(50°C) (same-sample)	—	14 mPa·s

Consolidating Figures



Consolidating figure: 5 diagnostic/forward-model panels (Γ-vs-q^2, apparent $R_h(q)$, static I(q) regime, $R_h \propto q\hat{}-\alpha$, CORENN PSD) plus the same-sample η(50 °C) solve (raw-derived).

Primary Model Results

q = 4πn sin(θ/2)/λ (n=1.36, λ=632.8 nm); the exported 'Scattering vector [1/nm]' column is mislabelled (its values are actually in 1/m), so q is recomputed from the angle and reported in nm^{-1}, spanning **0.00353–0.0261 nm^{-1}**. The size distribution is the **forward model** (CORENN multi-angle); Γ(q), apparent R_h(q), and the static I(q) regime are diagnostic projections.

22C (T = 295.1 K, recorded η = 34 mPa·s)

Step 2 — per-angle cumulants → Γ(q). A 2nd-order cumulant on the initial decay (g1_min=0.6) gives the decay rate at each angle; 28 angles are kept.

Step 3 — Γ vs q^2 (apparent diffusion). A straight-line fit Γ = c + D·q^2 over all 28 angles (no high-q rolloff detected) gives the **apparent diffusion coefficient D = 2.653e-14 ± 1.8e-16 m^2/s → R_h = 240 nm** (r^2=0.987) [replicate/repeatability 68% 210–214; angle/model 68% 206–217 nm]. The fitted intercept c ≈ -0/s is non-zero — a low-q curvature / non-diffusive offset (large-aggregate slow mode, baseline, or weak S(q)) — so read D as the slope of the linear region, not Γ/q^2. **The fit residuals are structured, not random** (figure inset): they trend systematically with q (Spearman resid–q = +0.49; runs 7 vs ˜14 expected) and a q^4 curvature term is significant at **+20σ** — so even with the rolloff removed, Γ vs q^2 genuinely **curves upward** — D_{app} rises with q, the signature of **polydispersity**: the line is a chord and no single slope is *the* diffusion coefficient. The q-dependent apparent size (next step) and the CORENN PSD are the proper description.

Step 4 — apparent R_h(q) (polydispersity). apparent R_h falls with q, from **506 nm** (low q) to **215 nm** (high q) (curvature C=1.91e-15, Spearman -0.97); mean PDI 0.30 — a broad / multi-modal population (large-particle tail

at low q).

Step 5 — static intensity I(q), regime from the PSD. The CORENN PSD gives dominant scatterers of R ≈ 530 nm ($R_g \approx 410$ nm, sphere-derived from the PSD radius — no Guinier regime is reached, so this is a model estimate, not a measured R_g), so q·R_g spans 1.4–10.7 across the angles → **form-factor / Porod (q·R $\gtrsim \pi$ throughout)**. On the 28 point contiguous lower-q scaling window (q=0.00353–0.0261 nm^{-1}): I $\propto q^{-3.16}\pm0.05$: a shallow / poorly-resolved power law (r^2=0.993) (reported as diagnostic only).

Step 6 — dynamic q-dependence. The apparent size follows a power law $R_{\mathrm{h,app}} \propto$ qˆ-α with α=0.58 (r^2=0.96) — a compact descriptor of the breadth of the distribution ($\alpha\to0$ monodisperse, $\alpha\to1$ fully scale-free).

Step 7 — CORENN multi-angle PSD (forward model). The joint inversion of the resolvable angles, on a radius grid bounded by dynamic observability plus the static q-window (PSD grid 8–591 nm, upper bound set by dynamic) and the main-relaxation lag window (λ=4.3e+00), recovers one q-consistent intensity size distribution: peak(s) at **530 nm (100%)** — this peak sits at the PSD upper observability bound (˜591 nm); the true material may extend beyond the measured lag/q window, so treat it as an unresolved slow-tail / lower-bound signature rather than a resolved peak.. An 20-angle bootstrap of the inversion puts the primary peak at 522 nm [68% 424–538; 95% 324–567 nm].

50C (T = 323.2 K, recorded η = 34 mPa·s)

Step 2 — per-angle cumulants → Γ(q). A 2nd-order cumulant on the initial decay (g1_min=0.6) gives the decay rate at each angle; 28 angles are kept.

Step 3 — Γ vs q^2 (apparent diffusion). A straight-line fit Γ = c + D·q^2 over all 28 angles (no high-q rolloff detected) gives the **apparent diffusion coefficient D = 8.093e-14 ± 4.8e-16 m^2/s → R_h = 86 nm** (r^2=0.935) [replicate/repeatability 68% 84–87; angle/model 68% 81–89 nm]. The fitted intercept c ≈ -2/s is non-zero — a low-q curvature / non-diffusive offset (large-aggregate slow mode, baseline, or weak S(q)) — so read D as the slope of the linear region, not Γ/q^2. **The fit residuals are structured, not random** (figure inset): they trend systematically with q (Spearman resid–q = +0.05; runs 3 vs ˜12 expected) and a q^4 curvature term is significant at **+21σ** — so even with the rolloff removed, Γ vs q^2 genuinely **curves upward** — D_{app} rises with q, the signature of **polydispersity**: the line is a chord and no single slope is *the* diffusion coefficient. The q-dependent apparent size (next step) and the CORENN PSD are the proper description.

Step 4 — apparent R_h(q) (polydispersity). apparent R_h falls with q, from **222 nm** (low q) to **84 nm** (high q) (curvature C=2.34e-15, Spearman -0.86); mean PDI 0.68 — a broad / multi-modal population (large-particle tail at low q).

Step 5 — static intensity I(q), regime from the PSD. The CORENN PSD gives dominant scatterers of R ≈ 203 nm ($R_g \approx 157$ nm, sphere-derived from the PSD radius — no Guinier regime is reached, so this is a model estimate, not a measured R_g), so q·R_g spans 0.6–4.1 across the angles → **spans Guinier (low-q) and Porod (high-q)**. A Guinier fit on the 5 low-q points (q·R_g < 1.3) gives $R_g \approx$ **366 nm** (r^2=0.94). On the 16 point contiguous lower-q scaling window (q=0.00353–0.0191 nm^{-1}): I $\propto q^{-3.12}\pm0.04$: a shallow / poorly-resolved power law (r^2=0.998) (reported as diagnostic only). The 12 higher-q point(s) are excluded from the exponent because they show systematic departure relative to the lower-q scaling law; physically this is the end of the mass-fractal / form-factor scaling range and/or an additive static background / instrument-floor contribution, not a second Porod law.

Step 6 — dynamic q-dependence. The apparent size follows a power law $R_{\mathrm{h,app}} \propto$ qˆ-α with α=0.71 (r^2=0.69) — a compact descriptor of the breadth of the distribution ($\alpha\to0$ monodisperse, $\alpha\to1$ fully scale-free).

Step 7 — CORENN multi-angle PSD (forward model). The joint inversion of the resolvable angles, on a radius grid bounded by dynamic observability plus the static q-window (PSD grid 8–234 nm, upper bound set by dynamic) and the main-relaxation lag window (λ=1.0e-06), recovers one q-consistent intensity size distribution: peak(s) at **203**

nm (100%). An 20-angle bootstrap of the inversion puts the primary peak at 193 nm [68% 145–208; 95% 98–213 nm].

Static Regime And Shape

The CORENN PSD anticipates which static regime each angle probes (Guinier where q·R_g < 1.3, Porod where q·R $\gtrsim \pi$). When a **valid Porod slope** exists, its exponent informs the form-factor / morphology model (q^{-4} a smooth sphere; 3<p<4 a surface fractal, D_s=6-p; p≈3 a rough-surface or branched network; 2≤p<3 a mass-fractal / branched aggregate, D=p; p≈2 a coil; p≈1 a rod). $\rho = R_g/R_h$ is the model-independent shape check (sphere ≈ 0.78, coil ≈ 1.5, rod $\gtrsim$ 2), determined only when a static Guinier R_g exists.

condition	q·R_g span	regime	static fit	implied morphology	$\rho = R_g/R_h$
22C	1.4–10.7	form-factor / Porod (q·R $\gtrsim \pi$ throughout)	I∝$q^{-3.16}$±0.05 (diagnostic only, q=0.00353–0.0261 nm^{-1})	—	not determined
50C	0.6–4.1	spans Guinier (low-q) and Porod (high-q)	Guinier R_g=366 nm	—	4.31 (rod / stiff / very extended (>1.9))

Particle model. The PSD assumes a homogeneous-sphere form factor. Where ρ is determined, compare it with the assumed model (sphere ≈ 0.78) and flag departures.

Error Propagation And Sensitivity

Headline R_h and the PSD peak carry bootstrap intervals (ρ where determined). The **replicate** bootstrap is measurement repeatability; the **angle/model** bootstrap is robustness to angle selection — the dominant term for polydisperse samples.

condition	R_h replicate 68%	R_h angle/model 68%	PSD peak 68%	ρ 68%
22C	210–214 nm	206–217 nm	424–538 nm	—
50C	84–87 nm	81–89 nm	145–208 nm	3.72–5.21

Systematic sensitivity (not included in the intervals above). $R_h = kT/(6\pi\eta D)$ scales inversely with viscosity, so a viscosity error propagates linearly into R_h; the q calibration (scattering angle + refractive index) shifts every q-dependent quantity; the absolute-intensity (toluene) scale sets the SLS magnitude; and the assumed sphere form factor biases the PSD radii for non-spherical particles. See the Assumption Register.

Fit Quality And Model Adequacy

Beyond r^2: a hold-out cross-validation (predictive R^2), a residual-structure test (runs / lag-1 autocorrelation — structured residuals flag model inadequacy even at high r^2), and the CORENN inversion's convergence and conditioning.

condition	Γ-vs-q^2 R^2 (in / CV)	Γ resid	CORENN g2 resid	CORENN: conv / GCV / κ / eff-DOF
22C	0.987 / 0.985	**structured**	**structured**	yes / interior / κ=1e+14 / 19/24
50C	0.935 / 0.926	**structured**	**structured**	yes / boundary / κ=5e+09 / 10/24

Structured Γ-vs-q^2 residuals reflect the known polydispersity curvature (the single slope is a summary, not the model); structured CORENN g2 residuals flag form-factor / single-distribution imperfection. A predictive CV R^2 well below the in-sample R^2 indicates over-fitting / poor extrapolation across angles.

Conditioning: $\kappa \approx$ 1e+14 means the inversion is severely ill-conditioned (˜14 of ˜16 double-precision digits lost), so the recovered distribution is fixed by the regularization, not the data alone — reinforcing the modality caveat. The eff-DOF/n_{basis} above is the multi-angle inversion's data-supportable complexity; it is a different inversion from the single-angle stitched one whose eff-DOF (˜5–6) appears in the size_max diagnostic.

Sample Synthesis

What the independent observables imply about the sample as a colloid — derived only from the scattering data above, with the inversion caveats carried through:

Physical picture. The scattering is dominated by **˜591 nm** objects (intensity-weighted); by number the population centres near **˜8 nm**, and the z-average R_h is 212 nm — best read as one broad, polydisperse distribution spanning these scales rather than well-separated species (see the size-distribution-by-weighting and size_max sections).

Hydrodynamic shape (Guinier–Porod → Stokes–Einstein). The static slope identifies the dominant scatterer as **a sphere (assumed)**, so the geometry-matched Stokes–Einstein relation is $\mathbf{D = kT/(6\pi\eta R)}$ — the radius R = R_h is **240 nm**. No resolved Porod slope, so the sphere form is assumed.

Not determinable from this angular range. The lowest q still has $q \cdot R_g \approx 1.4$ (> 1.3), so the **Guinier regime is never reached**: an independent radius of gyration, an absolute molar mass / aggregation number, and $\rho = R_g/R_h$ cannot be measured here — only inferred. Reaching them needs $q \lesssim 0.00317$ nm^{-1} (smaller scattering angles, low-q SLS, or SAXS).

Caveats that bound these conclusions. *(1) Dilute limit / S(q):* the analysis assumes single-particle scattering; if the suspension is not dilute, a low-q structure factor (interparticle correlations or incipient gelation) can bias the Porod/fractal slope and the apparent sizes — a dilution series would test it. *(2) Inversion dependence:* the modes and the aggregation number inherit the ill-posedness and size_max / λ / form-factor sensitivity documented in the size-distribution sections. *(3) Stability:* this is a single time-point of a system that can aggregate or sediment; a repeat / time-resolved run (or checking the count rate for drift across the angle scan) is needed to establish the distribution is stationary.

Consistency across observables. The broad dynamic distribution (mean PDI 0.30, apparent R_h falling with q) matches the polydisperse population inferred from the static slope. No single observable here is contradicted by another — the picture is one broad, polydisperse population rather than a set of distinct species.

Mode-Specific Diagnostics

Joint multi-temperature (same sample)

The recorded viscosity is a fixed 34 mPa·s at **both** temperatures — wrong for an electrolyte. The static intensity profile is unchanged with T (power-law exponent 3.2 at 22 °C vs 3.1 at 50 °C), supporting one (T-invariant) scattering structure; the 50 °C diffusion then *implies* the true viscosity:

- D_z(22°C)=2.998e-14, D_z(50°C)=8.209e-14 m^2/s; shared $R_{\text{h,z}}$=212 nm.
- **implied η(50 °C) = 14 mPa·s** (vs recorded 34), ratio $\eta(50)/\eta(22)$=0.40 — the electrolyte thins on heating.

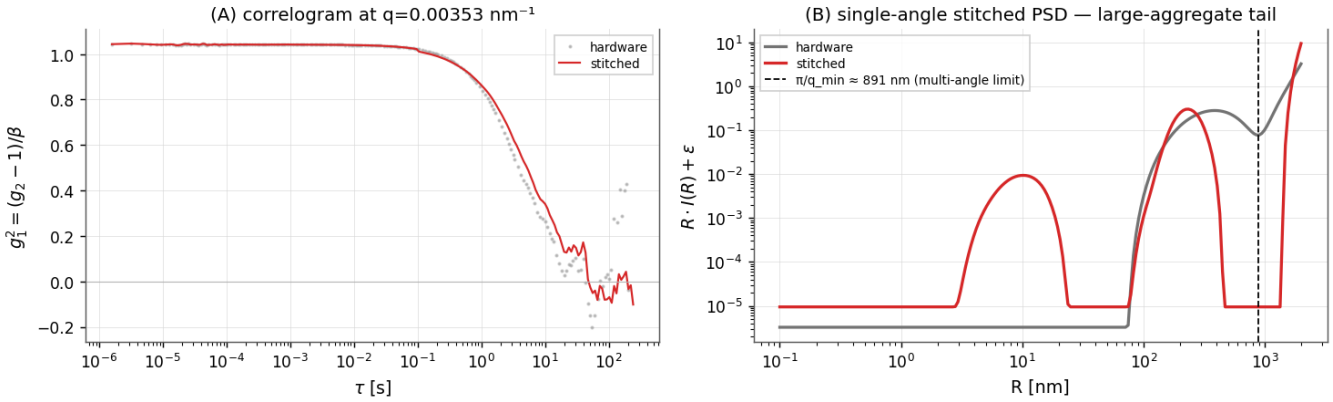
This is a cross-check (dynamics alone cannot separate η(T) from a size change); it is valid only because the *static* R_g is T-invariant.

Slow-mode / large-aggregate tail (single-angle stitched)

The multi-angle intensity PSD is q-resolution-limited: its R^6-weighted q→0 representation, coupled through the sphere form factor, edge-piles for $R \gtrsim \pi/q_{\min} \approx$ **891 nm**. A *single-angle* CORENN inversion of the gap-aware **stitched**

correlogram (clean long-lag pooled over the full run, τ_max 201→241 s) instead recovers the slow diffusive modes directly from the decay-rate spectrum — with a data-bounded amplitude, so it resolves the large-aggregate tail where the multi-angle PSD cannot.

Slow-mode / large-aggregate tail (single-angle stitched) — LiTFSI 22 °C

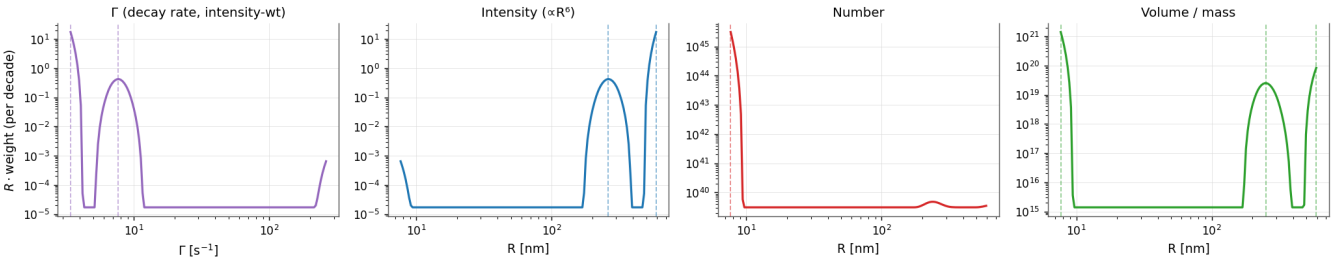


At the lowest angle (q=0.00353 nm^{-1}) the stitched inversion places **82% of the intensity beyond $\pi/q_{\min}$**, but it **piles at the radius-grid edge (~2 μm)**: the aggregates here are at/beyond even single-angle resolution, so this is a **lower bound** on a large-aggregate population (a genuinely slow diffusive mode), not a measured size — consistent with this sample sitting at the q-resolution limit.

Size distribution by weighting — continuous summary, then apparent modes

DLS recovers a size distribution by an **ill-posed Laplace inversion**: many different distributions — a smooth continuum or a set of discrete modes — reproduce the measured correlation function equally well within noise, and the regularized inversion (CORENN) returns just one of them, fixed by its smoothness penalty, radius grid, and form-factor assumption. The number and sharpness of peaks are therefore properties of the inversion, not facts the data establish on their own. The report gives **two views**: (1) assumption-light continuous percentiles, and (2) the inversion's apparent modes/shoulders, flagged as inversion-dependent and trustworthy only where they survive the size_max / λ sensitivity tests. CORENN returns the *same* inversion under every weighting (the vendor's Γ / Intensity / Number / Volume tabs); re-weighting only relocates and re-weights the features.

CORENN size distribution by weighting - LiTFSI 22 °C (compact-sphere conversions) -- dashed = peaks, dotted = shoulders



Γ (decay-rate) view — the rawest signal (figure panel 1). Before any size model, the dominant intensity component relaxes at $\Gamma \approx$ **3.41 s^{-1}** (relaxation time $\tau = 1/\Gamma \approx$ **293 ms**) at the reference angle; the size columns below are this spectrum mapped through Stokes–Einstein $\Gamma = k_B T\ q^2/(6\pi\eta R)$.

Continuous-distribution summary (percentiles; no modes assumed). The assumption-light description — percentiles of the recovered distribution, robust to how it is partitioned into modes. Intensity is narrow (dominated by the largest scatterers); number-weighting ($\div R^6$) strongly amplifies the inverted PSD's small-R end, so its percentiles are the least certain.

Weighting	D10 (nm)	D50 (nm)	D90 (nm)	D_g (nm)	σ_g (GSD)	span	log-skew
Intensity ($\propto R^6$)	287	568	587	514	1.31	0.53	-2.34
Number (sphere, R^6)	7.7	7.9	8.3	7.9	1.03	0.08	+5.42
Volume (sphere, R^3)	7.7	8.0	256	14	3.92	31.06	+1.98

Modality caveat — read the next table as inversion fine structure. The Laplace inversion is ill-posed: a smooth continuum and a set of discrete modes can fit the same g_2 within noise, so the number and sharpness of peaks depend on the regularization (λ), the radius grid / size_max, the spline basis, and the sphere form factor — none of which the data uniquely fix. The inversion carries ≈ **19 spline degrees of freedom** — ample latitude to place peaks on a continuum. **Treat the apparent peaks/shoulders below as fine structure of this particular inversion — not established sub-populations — unless they are stable under the size_max / λ sensitivity checks and reproducible across replicates.**

Apparent modes & shoulders (CORENN fine structure — inversion-dependent)

Weighting	Comp.	Peak R (nm)	⟨R⟩ (nm)	Median (nm)	Fraction	GSD	log-skew
Intensity ($\propto R^6$)	1 (edge)	≥591	570	572	87%	1.04	-0.92 (tail → small R)
	2	262	264	262	13%	1.13	-0.04 (≈ log-symmetric)
Number (sphere, R^6)	1 (edge)	≤7.7	7.9	7.9	100%	1.03	+1.03 (tail → large R)
Volume (sphere, R^3)	1 (edge)	≤7.7	8.0	7.9	85%	1.03	+0.95 (tail → large R)
	2	250	253	251	10%	1.13	+0.07 (≈ log-symmetric)
	3 (edge)	≥591	567	570	6%	1.04	-0.84 (tail → small R)

An edge component piles against the radius-grid bound (≤ small-R or ≥ large-R edge) — it is unresolved, so its size is a bound only and its ⟨R⟩/median/skew are not reliable. This is typical when number/volume weighting ($\div R^6$, $\div R^3$) amplifies the inverted PSD's small-R floor.

The intensity distribution is led by a peak at **591 nm** (87% of scattered light, tail → small R); under number weighting the dominant component is at **7.7 nm** — the *same* data, modes relocated and re-weighted by the R-power, which is why no single mean describes it.

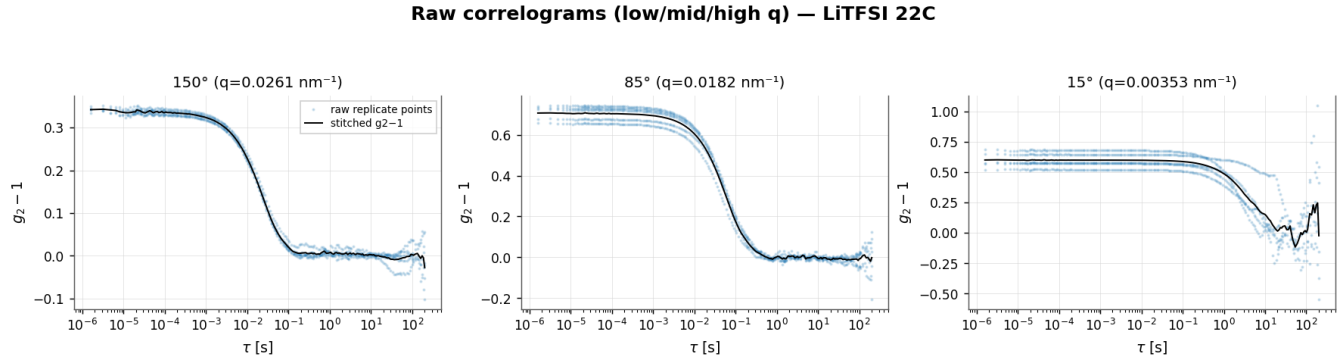
Volume/Number assume compact spheres ($V \propto R^3$); no mass-fractal correction is applied (the static slope is consistent with a compact / surface-fractal object, $D_f \approx 3$).

Data Inventory

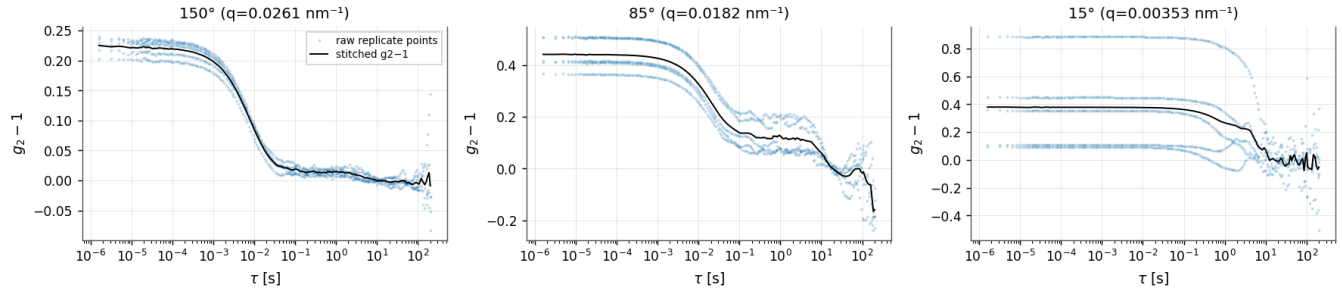
One row per analysed condition (per-angle detail is in Per-Curve Fit Results).

condition	channel	n angles (kept/total)	angle range	q range [nm^{-1}]	T [K]	scaled count rate [kHz/mW]	reps	missing meta
22C	3D	28/28	15–150°	0.00353–0.0261	295.1	0–76	28	0
50C	3D	28/28	15–150°	0.00353–0.0261	323.2	0–53	28	0

Raw Correlograms

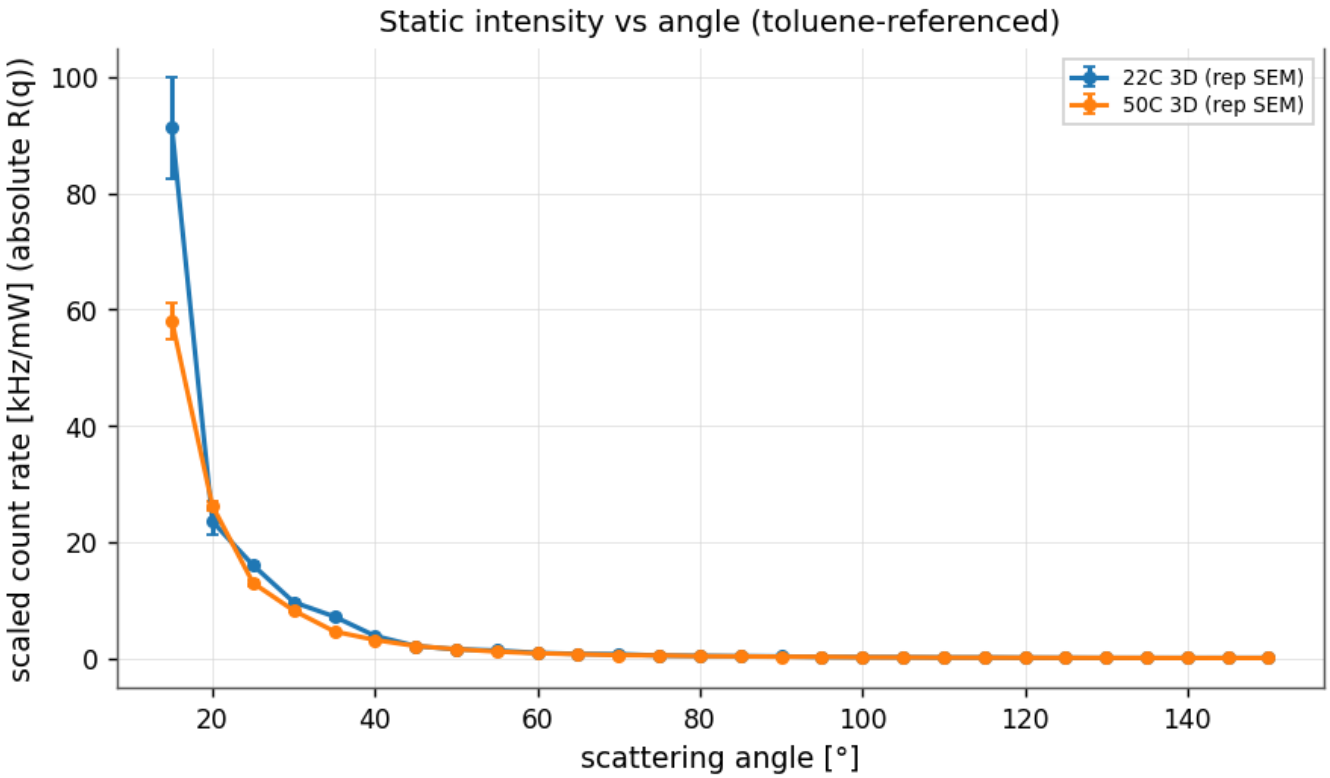


Raw 3D-cross correlograms at low/mid/high q (22C): the stitched multi-tau correlation (black) over the semi-transparent raw replicate points; the point spread is the measurement repeatability.



Raw 3D-cross correlograms at low/mid/high q (50C): the stitched multi-tau correlation (black) over the semi-transparent raw replicate points; the point spread is the measurement repeatability.

Count Rates And Static Intensities (absolute, toluene-referenced)



Scaled count rate (absolute Rayleigh ratio) vs angle for both temperatures (raw data).

Per-Curve Fit Results

Per-curve fits — 22C (3D)

Per-angle 2nd-order cumulant on the leading decay (g1_min=0.6); $R_{\mathrm{h,app}} = \mathrm{kT}\ \mathrm{q}^2/(6\pi\eta\ \Gamma)$.

angle [°]	q [nm^{-1}]	Γ [1/s]	μ_2	PDI	β	$R_{\mathrm{h,app}}$ [nm]	reps	accepted
15	0.00353	0.1	8.7e-03	0.71	0.576	713	5	yes
20	0.00469	0.3	4.5e-02	0.58	0.591	502	5	yes
25	0.00585	0.4	4.2e-02	0.26	0.654	543	5	yes
30	0.00699	0.7	3.0e-02	0.07	0.865	476	5	yes
35	0.00812	0.8	7.1e-02	0.10	0.875	506	5	yes
40	0.00924	1.2	1.4e-01	0.09	0.824	451	5	yes
45	0.0103	1.9	8.2e-01	0.23	0.817	361	5	yes
50	0.0114	2.5	8.5e-01	0.13	0.716	329	5	yes
55	0.0125	3.0	3.7e+00	0.42	0.818	333	5	yes
60	0.0135	3.8	4.3e+00	0.30	0.754	306	5	yes
65	0.0145	4.8	2.7e+00	0.11	0.747	277	5	yes
70	0.0155	5.3	7.8e+00	0.28	0.795	289	5	yes
75	0.0164	6.6	8.1e+00	0.19	0.742	260	5	yes
80	0.0174	7.0	1.2e+01	0.24	0.766	273	5	yes
85	0.0182	7.7	1.4e+01	0.23	0.675	275	5	yes
90	0.0191	8.7	2.6e+01	0.34	0.727	267	5	yes
95	0.0199	10.4	2.7e+01	0.26	0.648	243	5	yes
100	0.0207	10.7	3.4e+01	0.30	0.635	254	5	yes
105	0.0214	11.7	4.5e+01	0.33	0.650	250	5	yes
110	0.0221	12.7	6.7e+01	0.42	0.613	245	5	yes
115	0.0228	12.5	7.5e+01	0.48	0.572	265	5	yes
120	0.0234	15.4	9.3e+01	0.39	0.536	225	5	yes
125	0.024	15.5	7.7e+01	0.32	0.540	236	5	yes
130	0.0245	17.9	1.0e+02	0.31	0.486	213	5	yes
135	0.025	18.4	1.1e+02	0.33	0.492	215	5	yes
140	0.0254	18.9	1.5e+02	0.42	0.511	216	5	yes
145	0.0258	18.9	1.2e+02	0.33	0.399	223	5	yes
150	0.0261	20.3	1.3e+02	0.31	0.340	213	5	yes

Per-curve fits — 50C (3D)

Per-angle 2nd-order cumulant on the leading decay (g1_min=0.6); $R_{\mathrm{h,app}} = \mathrm{kT}\ \mathrm{q}^2/(6\pi\eta\ \Gamma)$.

angle [°]	q [nm^{-1}]	Γ [1/s]	μ_2	PDI	β	$R_{\mathrm{h,app}}$ [nm]	reps	accepted
15	0.00353	0.1	4.0e-03	0.37	0.094	833	5	yes
20	0.00469	0.6	4.0e-01	1.00	0.321	243	5	yes
25	0.00585	1.2	6.3e-01	0.42	0.256	195	5	yes
30	0.00699	1.5	1.9e+00	0.79	0.413	222	5	yes
35	0.00812	2.7	3.0e+00	0.43	0.352	173	5	yes
40	0.00924	4.5	5.7e+00	0.28	0.410	131	5	yes
45	0.0103	5.0	1.6e+01	0.64	0.437	150	5	yes
50	0.0114	5.9	2.1e+01	0.63	0.452	155	5	yes
55	0.0125	8.8	5.0e+01	0.65	0.489	123	5	yes
60	0.0135	10.1	7.7e+01	0.76	0.650	126	5	yes
65	0.0145	11.0	8.3e+01	0.69	0.449	133	5	yes
70	0.0155	13.9	1.2e+02	0.60	0.408	120	5	yes
75	0.0164	12.6	1.4e+02	0.86	0.404	149	5	yes
80	0.0174	18.0	2.3e+02	0.70	0.327	116	5	yes
85	0.0182	11.1	9.0e+01	0.74	0.365	210	5	yes
90	0.0191	18.1	2.8e+02	0.86	0.438	140	5	yes
95	0.0199	13.9	3.9e+02	2.03	1.345	199	5	yes
100	0.0207	30.1	5.7e+02	0.63	0.379	99	5	yes
105	0.0214	25.8	5.3e+02	0.80	0.320	124	5	yes
110	0.0221	36.2	7.9e+02	0.60	0.364	94	5	yes
115	0.0228	36.6	1.0e+03	0.77	0.409	99	5	yes
120	0.0234	40.6	9.4e+02	0.57	0.379	94	5	yes
125	0.024	44.2	9.8e+02	0.50	0.319	90	5	yes
130	0.0245	42.9	1.3e+03	0.69	0.308	97	5	yes
135	0.025	50.1	1.3e+03	0.52	0.285	87	5	yes
140	0.0254	54.0	1.6e+03	0.55	0.265	83	5	yes
145	0.0258	55.5	1.7e+03	0.55	0.237	83	5	yes
150	0.0261	56.7	1.7e+03	0.54	0.228	84	5	yes

Rejected Data And Failure Diagnostics

No angles or curves were rejected.

Assumption Register

assumption	value	source	affects	sensitivity checked?	comment
wavelength	632.8 nm	instrument default	q, D, R_h	no	—
refractive index	1.36	aqueous electrolyte, assumed	q	no	replace with measured
viscosity	34.00, 34.00 mPa·s	recorded export (fixed 34 mPa·s)	R_h	yes	fixed at both temperatures — placeholder, corrected by same-sample solve
q source	recalculated from angle	pipeline	all q-dependent fits	no	vendor 'Scattering vector [1/nm]' column is mislabelled (values are 1/m); q recomputed from angle, reported in nm^{-1}
static scale	absolute (toluene-referenced) — assumed, unverified	LSII toluene reference	SLS, R_g	no	toluene reference not confirmable from parsed export
form factor	homogeneous sphere (RGD)	pipeline	PSD radii, ρ check	yes	flag if $\rho \neq 0.78$ or depolarization present
cumulant g1_min	0.6	pipeline config	Γ, PDI	yes	leading-decay window
PSD radius grid	8–591 nm	dynamic observability + static q guardrail	PSD peaks	yes	upper=min(dynamic 591 nm, static 670 nm, default 5000 nm); limiting=dynamic; tau_max=0.94 s, min field decay=0.223
PSD basis	24 B-spline control points	pipeline config	PSD smoothness	yes	—
regularization	GCV-selected λ=4.3e+00	pipeline	PSD width	yes	interior minimum required
baseline	free per fit (CORENN); leading-decay cumulant	pipeline config	Γ, PSD	yes	—

Quality Gates

Provisional report: **11 PASS, 2 WAIVE, 0 FAIL** of 13.

gate	status	note
q physically plausible / reproducible from angle	**PASS**	q recomputed from angle (vendor q column mislabeled 1/nm)
viscosity not a placeholder	**WAIVE**	same η recorded across temperatures — flagged placeholder
fit bounds stated for every fit	**PASS**	cumulant g1_min=0.6; CORENN tau_max=0.94s, Rmax=591 nm (dynamic)
rejected curves/angles listed	**PASS**	0 rejected, tabulated
residual-structure test reported (not r^2 alone)	**PASS**	runs test + lag-1 autocorrelation
headline R_h carries a bootstrap CI	**PASS**	replicate + angle/model bootstrap
hold-out cross-validation reported	**PASS**	leave-one-angle-out
CORENN converged + well-posed regularization	**PASS**	converged=True, GCV interior=True, κ=1e+14
static scale confirmed absolute	**WAIVE**	toluene reference assumed (LSII default), not confirmable from parsed export
PSD basis (intensity/volume/number) stated	**PASS**	intensity-weighted primary, conversions noted
$\rho = R_g/R_h$ and particle model stated	**PASS**	not determined; sphere form factor assumed
multi-temperature size/viscosity ambiguity addressed	**PASS**	same-sample joint-temperature solve reported
figure captions state raw vs derived	**PASS**	captions included

Metadata

field	value
Report title	LiTFSI 20 molal — provisional DLS/SLS analysis report
Data source	data/LiTFSI/20m
Report generator	reports/litfsi_report.py (provisional)
Analyst / owner	automated pipeline
Dataset class	multi-angle, multi-temperature (3D cross-correlation)
Channels present	3D
Temperatures	295.1 K, 323.2 K
Angles	15–150° (56 per condition)
q range	0.00353–0.0261 nm^{-1}
Replicates	56 angle-sets with ≥2 replicates
Solvent constants	η per condition (see table); n=1.36, λ=632.8 nm (aqueous electrolyte, assumed)
Static scale	absolute (toluene-referenced) — assumed, unverified
Particle model	homogeneous sphere (assumed; ρ not determined to confirm)
Primary model	CORENN multi-angle PSD + same-sample joint-temperature solve
Report limitations	viscosity placeholder; toluene reference unverified; sphere PSD form factor inconsistent with the Porod slope

Reproducibility Appendix

item	value
Command	`python reports/litfsi_report.py`
Environment	python 3.13.5, numpy 2.2.5
Git commit (working tree)	91a3b68 (dirty)
Random seeds	bootstraps seed=0 (fixed)
Generated files	LiTFSI.md, LiTFSI.pdf, figures/litfsi.png, figures/litfsi_slowmode.png, figures/litfsi_weightings.png, figures/litfsi_raw_22C.png, figures/litfsi_raw_50C.png, figures/litfsi_countrate.png, LiTFSI.sidecar.json